

# SAFETY DATA SHEET



Revision date 12-Dec-2024

Revision Number 1.2

## 1. Identification

### Product identifier

**Product Name** Salmonella typhi - Antigen O (TO), 50 mL

### Other means of identification

**Catalog Number(s)** 63402

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users In-vitro laboratory reagent or component

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com\*\*\*

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Clinical Diagnostics  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA\*\*\*

**Technical Service** 1-800-224-6723  
TechSupport.USSD@bio-rad.com\*\*\*

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                    |                |
|--------------------|----------------|
| Skin sensitization | Category 1***  |
| Carcinogenicity    | Category 1A*** |

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

**Danger\*\*\***

#### Hazard statements

May cause an allergic skin reaction  
May cause cancer\*\*\*



|                          |                              |                 |
|--------------------------|------------------------------|-----------------|
| <b>Appearance</b> Liquid | <b>Physical state</b> Liquid | <b>Odor</b> Low |
|--------------------------|------------------------------|-----------------|

**Precautionary Statements - Prevention**

Wear protective gloves/clothing and eye/face protection\*\*\*

**Precautionary Statements - Response**

IF exposed or concerned: Get medical advice/attention\*\*\*

IF ON SKIN: Wash with plenty of water and soap

If skin irritation or rash occurs: Get medical advice/attention\*\*\*

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable\*\*\*

**Other information**

Causes mild skin irritation.\*\*\*

Contains human source material and / or potentially infectious components\*\*\*

**3. Composition/information on ingredients****Substance**

Not applicable.\*\*\*

**Mixture** \*\*\*

| Chemical name | CAS No. | Weight-%    | Trade secret |
|---------------|---------|-------------|--------------|
| Formaldehyde  | 50-00-0 | 0.1 - 0.249 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

**4. First-aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance. IF exposed or concerned: Get medical advice/attention.\*\*\*

**Inhalation**

Remove to fresh air.

**Eye contact**

Contains human source material and / or potentially infectious components. Call a physician. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.\*\*\*

**Skin contact**

Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician.\*\*\*

**Ingestion**

Contains human source material and / or potentially infectious components. Call a physician.\*\*\*

**Most important symptoms and effects, both acute and delayed****Symptoms**

Itching. Rashes. Hives. Prolonged contact may cause redness and irritation.\*\*\*

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components.\*\*\*

**5. Fire-fighting measures**

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Specific hazards arising from the chemical** Product is or contains a sensitizer. May cause sensitization by skin contact.\*\*\*

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

**6. Accidental release measures****Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.\*\*\*

**Other information** Refer to protective measures listed in Sections 7 and 8.\*\*\*

**Methods and material for containment and cleaning up**

**Methods for containment** Do not allow into any sewer, on the ground or into any body of water.\*\*\*

**Methods for cleaning up** Use: Disinfectant. Clean contaminated surface thoroughly.\*\*\*

**7. Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.\*\*\*

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.\*\*\*

## 8. Exposure controls/personal protection

### Control parameters

#### Exposure Limits

\*\*\*

| Chemical name           | ACGIH TLV                                                                    | OSHA PEL                                                                                                                                                                                                                                        | NIOSH                                                     |
|-------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Formaldehyde<br>50-00-0 | dermal sensitizer;respiratory<br>sensitizer<br>STEL: 0.3 ppm<br>TWA: 0.1 ppm | TWA: 0.75 ppm<br>(vacated) TWA: 3 ppm<br>unless specified in 1910.1048<br>(vacated) STEL: 10 ppm 30<br>min unless specified in<br>1910.1048<br>(vacated) Ceiling: 5 ppm unless<br>specified in 1910.1048<br>STEL: 2 ppm see 29 CFR<br>1910.1048 | IDLH: 20 ppm<br>Ceiling: 0.1 ppm 15 min<br>TWA: 0.016 ppm |

### Appropriate engineering controls

#### Engineering controls

Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

#### Eye/face protection

Wear safety glasses with side shields (or goggles).\*\*\*

#### Hand protection

Wear suitable gloves.\*\*\*

#### Skin and body protection

Wear suitable protective clothing.\*\*\*

#### Respiratory protection

No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

#### General hygiene considerations

Do not eat, drink or smoke when using this product. Wash hands before breaks and immediately after handling the product. Follow universal and standard precautions for handling potentially infectious materials.\*\*\*

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical state | Liquid                   |
| Appearance     | Liquid                   |
| Color          | white                    |
| Odor           | Low                      |
| Odor threshold | No information available |

#### Property

#### Values

#### Remarks • Method

|                                         |                   |            |
|-----------------------------------------|-------------------|------------|
| pH                                      | No data available | None known |
| Melting point / freezing point          | No data available | None known |
| Initial boiling point and boiling range | No data available | None known |
| Flash point                             | No data available | None known |
| Evaporation rate                        | No data available | None known |
| Flammability                            | No data available | None known |
| Flammability Limit in Air               |                   | None known |

|                                        |                   |            |
|----------------------------------------|-------------------|------------|
| Upper flammability or explosive limits | No data available |            |
| Lower flammability or explosive limits | No data available |            |
| Vapor pressure                         | No data available | None known |
| Relative vapor density                 | No data available | None known |
| Relative density                       | No data available | None known |
| Water solubility                       | Miscible in water |            |
| Solubility(ies)                        | No data available | None known |
| Partition coefficient                  | No data available | None known |
| Autoignition temperature               | No data available | None known |
| Decomposition temperature              | No data available | None known |
| Kinematic viscosity                    | No data available | None known |
| Dynamic viscosity                      | No data available | None known |

**Other information**

|                      |                          |
|----------------------|--------------------------|
| Explosive properties | No information available |
| Oxidizing properties | No information available |
| Softening point      | No information available |
| Molecular weight     | No information available |
| VOC content          | No information available |
| Liquid Density       | No information available |
| Bulk density         | No information available |

**10. Stability and reactivity**

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

**11. Toxicological information****Information on likely routes of exposure**

|                     |                                                                                                                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product Information | ***                                                                                                                                                                                                                                                        |
| Inhalation          | Specific test data for the substance or mixture is not available.                                                                                                                                                                                          |
| Eye contact         | Specific test data for the substance or mixture is not available.                                                                                                                                                                                          |
| Skin contact        | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). Causes mild skin irritation.*** |
| Ingestion           | Specific test data for the substance or mixture is not available.                                                                                                                                                                                          |

**Symptoms related to the physical, chemical and toxicological characteristics**

|          |                                                                                |
|----------|--------------------------------------------------------------------------------|
| Symptoms | Itching. Rashes. Hives. Prolonged contact may cause redness and irritation.*** |
|----------|--------------------------------------------------------------------------------|

**Acute toxicity**

## Numerical measures of toxicity

\*\*\*

## Component Information

\*\*\*

| Chemical name           | Oral LD50           | Dermal LD50          | Inhalation LC50       |
|-------------------------|---------------------|----------------------|-----------------------|
| Formaldehyde<br>50-00-0 | = 100 mg/kg ( Rat ) | > 2000 mg/kg ( Rat ) | < 463 ppm ( Rat ) 4 h |

Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Classification based on data available for ingredients. Causes mild skin irritation.\*\*\*

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitization** May cause an allergic skin reaction.\*\*\*

**Germ cell mutagenicity** No information available.

**Carcinogenicity** Contains a known or suspected carcinogen. Classification based on data available for ingredients. May cause cancer.\*\*\*

The table below indicates whether each agency has listed any ingredient as a carcinogen.\*\*\*

| Chemical name           | ACGIH | IARC    | NTP   | OSHA |
|-------------------------|-------|---------|-------|------|
| Formaldehyde<br>50-00-0 | A1    | Group 1 | Known | X    |

## Legend

\*\*\*

**ACGIH (American Conference of Governmental Industrial Hygienists)**

A1 - Known Human Carcinogen\*\*\*

**IARC (International Agency for Research on Cancer)**

Group 1 - Carcinogenic to Humans\*\*\*

**NTP (National Toxicology Program)**

Known - Known Carcinogen\*\*\*

**Occupational Safety and Health Administration of the US Department of Labor**

X - Present\*\*\*

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

## 12. Ecological information

**Ecotoxicity** The environmental impact of this product has not been fully investigated.

| Chemical name           | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                 | Toxicity to microorganisms | Crustacea                                                                     |
|-------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------|
| Formaldehyde<br>50-00-0 | -                    | LC50: 22.6 - 25.7mg/L (96h, Pimephales promelas)<br>LC50: =1510µg/L (96h, Lepomis macrochirus)<br>LC50: =41mg/L (96h, Brachydanio rerio)<br>LC50: 0.032 - 0.226mL/L (96h, Oncorhynchus mykiss)<br>LC50: 100 - 136mg/L (96h, Oncorhynchus mykiss)<br>LC50: 23.2 - 29.7mg/L (96h, Pimephales promelas) | -                          | LC50: =2mg/L (48h, Daphnia magna)<br>EC50: 11.3 - 18mg/L (48h, Daphnia magna) |

**Persistence and degradability** No information available.

**Bioaccumulation** \*\*\*

**Component Information** \*\*\*

| Chemical name           | Partition coefficient |
|-------------------------|-----------------------|
| Formaldehyde<br>50-00-0 | 0.35                  |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.\*\*\*

**Contaminated packaging** Do not reuse empty containers.

**California Hazardous Waste Status** This product contains one or more substances that are listed with the State of California as a hazardous waste.\*\*\*

## 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

IATA Not regulated

IMDG Not regulated

## 15. Regulatory information

International Inventories Contact supplier for inventory compliance status

### US Federal Regulations

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.\*\*\*

| Chemical name          | SARA 313 - Threshold Values % |
|------------------------|-------------------------------|
| Formaldehyde - 50-00-0 | 0.1                           |

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

#### **CWA (Clean Water Act)**

This product contains the following substances which are regulated pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).\*\*\*

| Chemical name           | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants | CWA - Hazardous Substances |
|-------------------------|-----------------------------|------------------------|---------------------------|----------------------------|
| Formaldehyde<br>50-00-0 | 100 lb                      | -                      | -                         | X                          |

#### **CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).\*\*\*

| Chemical name           | Hazardous Substances RQs | Extremely Hazardous Substances RQs | Reportable Quantity (RQ)                  |
|-------------------------|--------------------------|------------------------------------|-------------------------------------------|
| Formaldehyde<br>50-00-0 | 100 lb                   | 100 lb                             | RQ 100 lb final RQ<br>RQ 45.4 kg final RQ |

### US State Regulations

#### **California Proposition 65**

This product contains the following Proposition 65 chemicals:\*\*\*

| Chemical name          | California Proposition 65 |
|------------------------|---------------------------|
| Formaldehyde - 50-00-0 | Carcinogen                |

#### **U.S. State Right-to-Know Regulations**

\*\*\*

| Chemical name           | New Jersey | Massachusetts | Pennsylvania |
|-------------------------|------------|---------------|--------------|
| Water<br>7732-18-5      | -          | -             | X            |
| Formaldehyde<br>50-00-0 | X          | X             | X            |

### U.S. EPA Label Information



EPA Pesticide Registration Number Not applicable

## 16. Other information

|                                   |                              |                       |                           |                              |
|-----------------------------------|------------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b>                       | <b>Health hazards</b> 2***   | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b>                       | <b>Health hazards</b> 2 **** | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X |
| <i>Chronic Hazard Star Legend</i> |                              |                       |                           |                              |
| * = Chronic Health Hazard***      |                              |                       |                           |                              |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision date** 12-Dec-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**